Week 12: 16 Nov 2026 –20 Nov 2026

Subject: Mathematics **Unit Title:** Algebra Foundations & Introduction to Polynomials **Grade :** 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative/تكويني Summative /ختامي Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: Functions	I can identify domain, range and intercepts of a function	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	Girl's campus: Materials: Teams channel, file section>Term 1 Boy's campus Materials: Teams channel Assignments are to be submitted as the teacher deadline.
Lesson 2: Functions	I can analyze the graph of a function.		
Lesson 3: Polynomial function	I can graph a polynomial function.		
Lesson 4: Quadratic function	I can graph and analyze quadratic functions.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	**Make sure your devices are fully charged.** **Bring your pencil case/calculator with you.** **Grade 10 will have an official assignment to be submitted weekly according to the teacher's announcement on Toddle.** **This week, the students need to complete the assignment sent on teams by the teacher.** **Homework: GRASPS TASK shared on Toddle, submission due Wednesday, 18 November.**		

Technology Integration دمج التكنولوجيا	Resources shared on Teams
Cross-curricular links الروابط بين المواد الدراسية	**Numeracy**: Students operate with polynomial expressions, solve quadratic equations using multiple algebraic methods, apply the Remainder Theorem, calculate geometric series sums, and compute and interpret average rates of change — all in authentic UAE-embedded contexts. **Critical & Creative Thinking:** Students analyze real-world polynomial models, evaluate the meaning of solutions (including complex ones), and make mathematical arguments about rates of change, optimization, and cumulative growth.

	Communication: Students write mathematical reports for a non-specialist UAE audience, presenting polynomial models and their interpretations clearly and precisely. GRASPS task presentations develop oral and written mathematical communication. **Self-management:** Students manage multi-session GRASPS task work across the unit, track their own progress using project journals and exit tickets, and reflect on their learning after the summative assessment.
External Assessment Integration دمج التقييمات الخارجية	All content is covered in MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Wellbeing:** • Confidence in solving real-world maths problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussions. • Self-awareness by setting and reflecting on learning goals. **IB ATL Skills:** • **Thinking:** solving problems, recognizing patterns, and applying proportional reasoning. • **Communication:** explaining mathematical thinking using appropriate vocabulary. • **Research:** investigating real-world examples and interpreting scale drawings. • **Self-Management:** setting goals, organizing work, and managing time effectively. • **Social:** collaborating respectfully and sharing ideas with others.
Standards المعايير	N.CN.7 Solve quadratic equations with real coefficients that have complex solutions. F.IF.8a Use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can recognizing patterns help us factor quadratic expressions efficiently? How can we transform a quadratic equation into a perfect square to find its solutions? How does the leading coefficient affect the process of completing the square? How can we represent quadratic solutions when they are not rational numbers?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1: ▨ All students must be able to find the range, domain of a function. ◍ Most students should be able to find the intercepts of a function. ▲ Some students could be able to graph a function. Lesson 2: ▨ All students must be able to analyse the graph of a function. ◍ Most students should be able to find the roots graphically.

	▲ Some students could be able to solve functions equations graphically by drawing the horizonal line of equation: y=a. Lesson 3: ▨ All students must be able to graph a polynomial function. ◍ Most students should be able to find the end behaviors of the polynomial function graphically. ▲ Some students could be able to solve polynomial functions equations graphically. Lesson 4: ▨ All students must be able to graph a quadratic function. ◍ Most students should be able to sketch the quadratic function graph when the sign of a changes. ▲ Some students could be able to solve real word problems about quadratic equations graphically.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Respect الاحترام Heritage التراث
	Students solve quadratic problems connected to **UAE development and infrastructure**, such as calculating dimensions, areas, and design measurements using factoring and completing the square Students demonstrate **respect for different mathematical approaches** by comparing factoring, completing the square, and exact radical solutions while discussing and justifying their strategies with classmates. Students apply quadratic concepts to **patterns and geometric designs inspired by UAE traditional architecture and heritage**, using factoring and quadratic equations to investigate dimensions and symmetry.